

**Mohammed Reabai J 250701435-2025-CSE** ▾**M2****Started on** Friday, 25 September 2026, 3:21 PM**State** Finished**Completed on** Friday, 25 September 2026, 3:21 PM**Time taken** 23 secs**Marks** 1.00/1.00**Grade** 10.00 out of 10.00 (100%)

Question 1 | Correct | Mark 1.00 out of 1.00

Find Duplicate in Array.

Given a read only array of n integers between 1 and n, find one number that repeats.

Input Format:

First Line - Number of elements

n Lines - n Elements

Output Format:

Element x - That is repeated

For example:

Input	Result
5 1 1 2 3 4	1

Answer: (penalty regime: 0 %)

```

1  #include <stdio.h>
2
3  int main() {
4      int n, i;
5      scanf("%d", &n);
6
7      int a[n];
8
9      for (i = 0; i < n; i++) {
10         scanf("%d", &a[i]);
11     }
12
13     int slow = a[0];
14     int fast = a[0];
15
16     do {
17         slow = a[slow];
18         fast = a[a[fast]];
19     } while (slow != fast);
20
21     slow = a[0];
22
23     while (slow != fast) {
24         slow = a[slow];
25         fast = a[fast];
26     }
27
28     printf("%d", slow);
29
30     return 0;
31 }
32

```

	Input	Expected	Got	
✓	11 10 9 7 6 5 1 2 3 8 4 7	7	7	✓
✓	5 1 2 3 4 4	4	4	✓
✓	5 1 1 2 3 4	1	1	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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